



Shreyansh Tiwari

Bachelor of Technology
in Computer Science and Engineering
National Institute of Technology, Kurukshetra

+91-7014508415
124102054@nitkkr.ac.in
GitHub
LinkedIn

SUMMARY

Computer Science undergraduate at NIT Kurukshetra with expertise in Data Structures & Algorithms, distributed backend systems, and full-stack web development. Skilled in designing scalable, low-latency applications using Node.js, Express.js, React.js, MongoDB, Redis, BullMQ, RAG and Gemini API. Strong competitive programming background with 1800+ solved problems and consistent performance in national and international programming contests. Passionate about solving complex engineering challenges, optimizing system performance, and building reliable software at scale.

EDUCATION

Degree	Institute	Board / University	CGPA/Percentage	Year
B.Tech CSE	National Institute of Technology, Kurukshetra	NIT Kurukshetra	9.1803	2024-2028
Senior Secondary	Shivam Public School	CBSE	82.8%	2022-2023
Matriculation	Shivam Public School	CBSE	94.00%	2020-2021

PROJECTS

- Portfolio Website with AI Assistant** July 2026 - Present
React.js, Node.js, Express.js, Gemini API, RAG, JavaScript **GitHub – Live**
 - Developed a full-stack portfolio website featuring an AI assistant powered by Retrieval-Augmented Generation (RAG) to answer questions about projects, skills, education, and experience.
 - Built a custom semantic search pipeline by parsing and chunking structured knowledge, generating vector embeddings, and retrieving relevant context using cosine similarity before LLM inference.
 - Designed a modular knowledge base using JSON documents and PDF parsing, enabling accurate retrieval across projects, achievements, resume, and technical skills without hardcoded responses.
- Chess Multiplayer Web App** Jan 2026 - Feb 2026
HTML, CSS, Node.js, Socket.io, MongoDB, JWT, REST API **GitHub – Live**
 - Architected and developed a real-time multiplayer chess platform supporting room-based matchmaking, concurrent game sessions, and synchronized gameplay across distributed clients.
 - Designed a server-authoritative game engine that enforces legal moves, turn management, check/checkmate detection, and consistent board-state synchronization using WebSocket communication.
 - Implemented secure JWT-based authentication, protected routes, and persistent user management with MongoDB, enabling authenticated multiplayer gameplay and session handling.

TECHNICAL SKILLS

- Programming Languages:** C, C++, Python, JavaScript, Java
- Web:** React.js, Node.js, Express.js, REST APIs, HTML, CSS, Tailwind CSS
- AI/ML:** scikit-learn, NumPy, Pandas, OpenCV, Retrieval-Augmented Generation (RAG)
- Databases:** MongoDB, Firebase, MySQL
- Tools/DevOps:** Docker, Git, GitHub, Postman, Kubernetes, AWS

ACHIEVEMENTS

- ICPC 2025 Preliminary Round**, Secured **AIR 473**.
- Competitive Programming Ratings**, **1999 Knight** @LeetCode, **Specialist (1470)** @CodeForces, **3 Star** @CodeChef.
- Top Global Contest Ranks**, CodeChef Starters 221 : **190**, LeetCode Biweekly 185: **252**, Codeforces Round 1084: **952**.
- Problem Solving**, Solved **1800+** Data Structures and Algorithms / Competitive Programming problems.
- Social Summer of Code (SSoC)** :, Open-source contributor; collaborated on real-world repositories and merged pull requests.

POSITIONS OF RESPONSIBILITY

- Technobyte – Technical Society, NIT Kurukshetra** 2024 – Present
Active Member
 - Contributing to technical workshops, coding contests, and collaborative learning initiatives.

KEY COURSES TAKEN

- Core Computer Science:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Computer Organization and Architecture
- Programming & Web:** Object-Oriented Programming, Problem Solving using C/C++, Web Development
- Mathematics:** Discrete Mathematics